JoyRobots Rangers

EAB Research Detectives

Week 6 | October 24, 2026 | Research homework

Ranger Name: _____

Your mission: Real scientists find facts, they don't just memorize them. Use the two websites below to collect clues about the emerald ash borer (EAB) for our Tree Stethoscope. Plan for about 30 minutes; an adult can help with tricky words. **Detective rule:** write answers in your own words.

Where to look (scan the QR code with a phone, or type the search words)

A Natural Resources Canada — Forest insects and diseases

Search for: NRCan forest insects and diseases

natural-resources.canada.ca/forests-forestry/forest-insects-diseases

Click **Identifying forest pests**, then find the card **Emerald ash borer (EAB)**. Click its title to open the full factsheet if it will open.

B Smith Lab (University of Toronto forest health lab) — Links page

Search for: smithlab.ca contact links

smithlab.ca/contact--links.html

This is a list of links, not an article. Under **Forest Health & Urban Forest Links**, pick **one** link, for example **Tree Killers in Canada's Forests** or **Ontario Forest Health**, and search that site for "emerald ash borer".

1. **[A]** How do EAB larvae hurt a tree? Hint: where do they live, and what do they cut off?

2. **[A]** Write two signs that a tree has EAB. Circle the one you could see from the sidewalk.

3. **[A]** How do people protect ash trees or slow EAB today? Name one way.

4. **[B]** Which link did you pick, and what is the title of the EAB page you found?

5. **[B]** Write one new fact that was <u>not</u> on website A.

Think like a Tree Stethoscope engineer

6. A teammate says: "Our sensor heard nothing, so this tree has no EAB." Give one reason the sensor could hear nothing even if larvae are inside. Hint: season, larva size, where the sensor sits, other noise.

Bring to our next meeting: this sheet, plus one surprising fact and one question your research could not answer.